

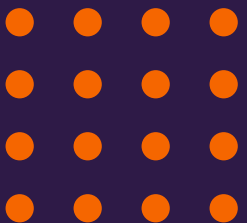


Optimization and Regularization of Neural Networks

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Outline

Optimization

Regularization of NN



Outline

Optimization

- Challenges in Optimization
- Momentum
- Adaptive Learning Rate
- Parameter Initialization
- Batch Normalization

Regularization of NN

- Norm Penalties
- Early Stopping
- Data Augmentation
- Sparse Representation
- Dropout



Learning vs. Optimization

Goal of learning: minimize generalization error, or the loss function

$$\mathcal{L}(W) = \mathbb{E}_{(x,y) \sim p_{data}} [L(f(x, W), y)]$$

In practice, empirical risk minimization:

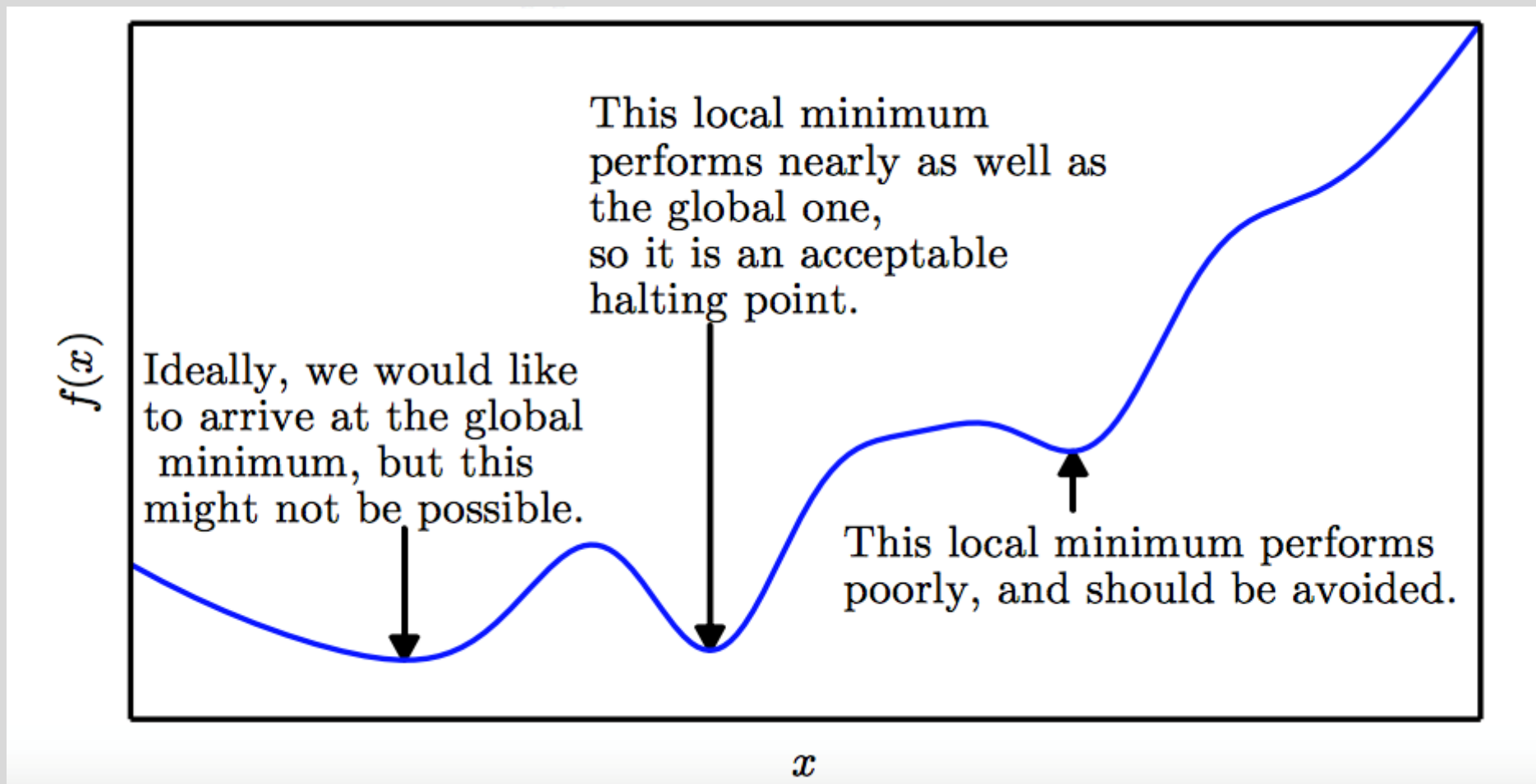
$$\mathcal{L}(W) = \sum_i [L(f(x_i; W), y_i)]$$

f is the neural network

Quantity optimized
different from the quantity
we care about



Local Minima





Critical Points

Points with zero gradient

2nd-derivate (Hessian) determines curvature

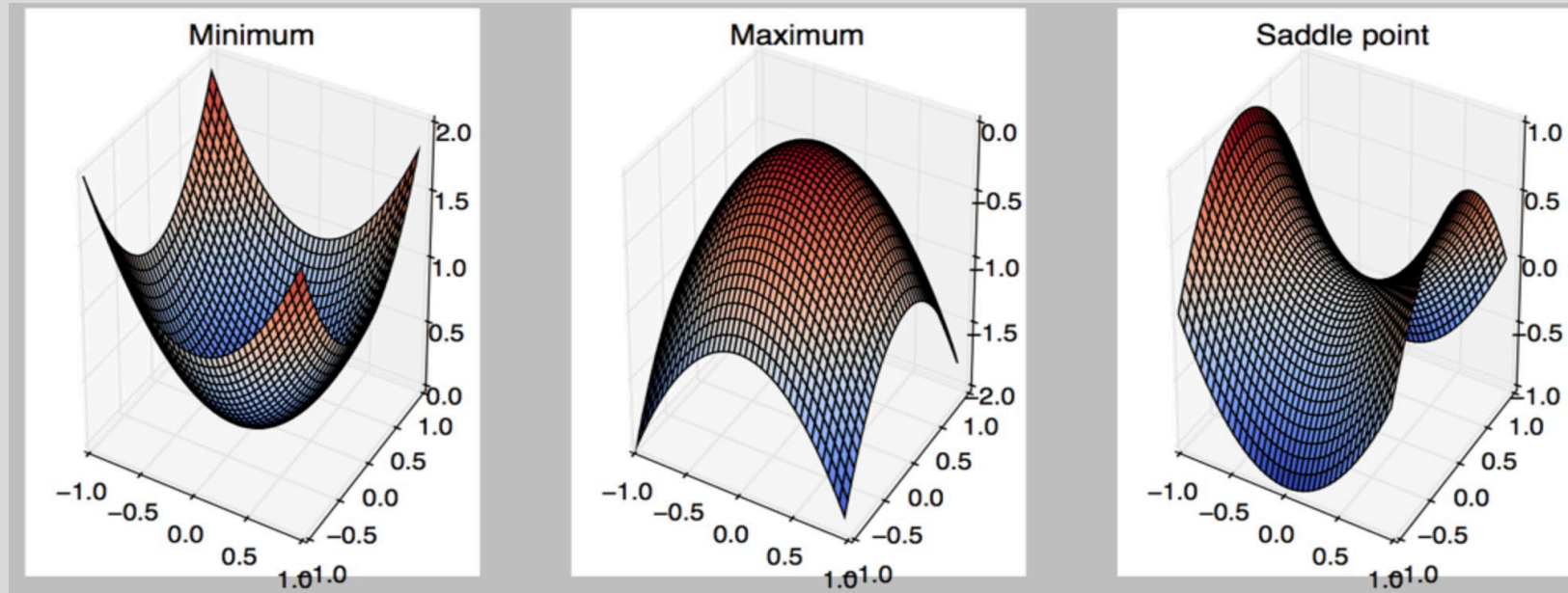


Image Source: Goodfellow et al. (2016)

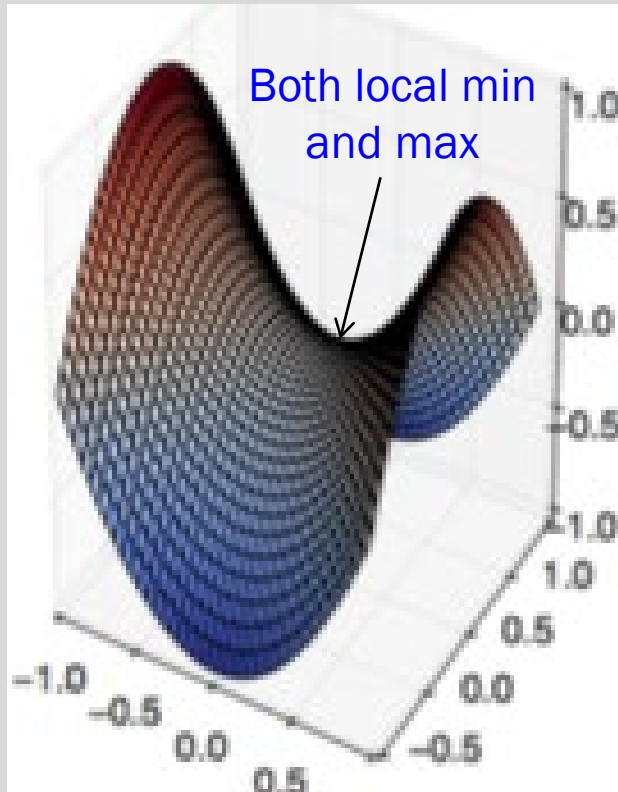


Local Minima

Old view: local minima is major problem in neural network training

Recent view:

- For sufficiently large neural networks, **most local minima incur low cost**
- Not important to find true global minimum



Saddle Points

Recent studies indicate that in high dim, saddle points are more likely than local min

Gradient can be very small near saddle points

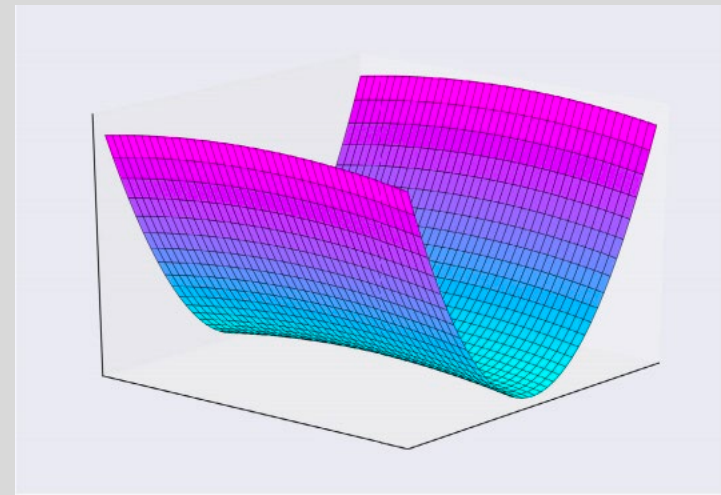


Image Source: Goodfellow et al. (2016)



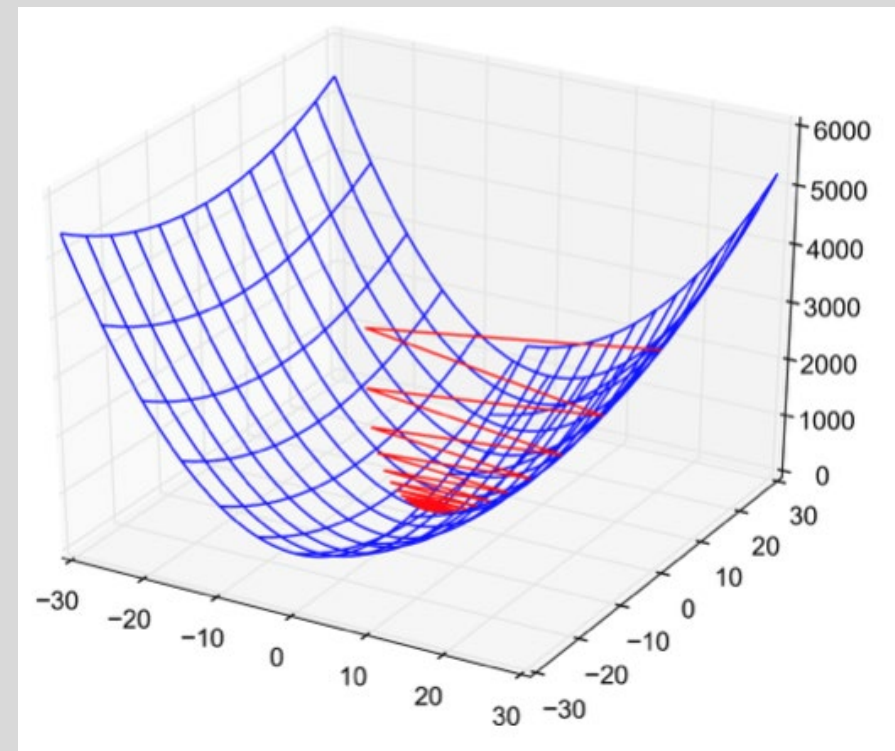
Poor Conditioning

Poorly conditioned Hessian matrix

- **High curvature:** small steps leads to huge increase

Learning is slow despite strong gradients

Oscillations slow down progress

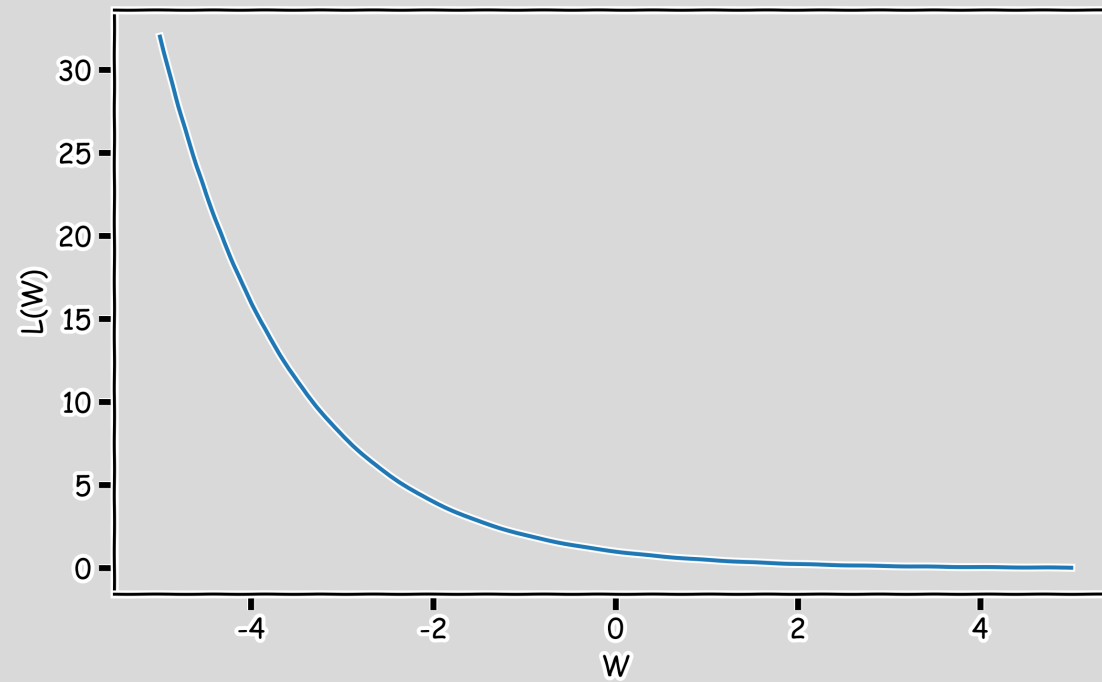




No Critical Points

Some cost functions do not have critical points. In particular: classification.

WHY?





Exploding and Vanishing Gradients

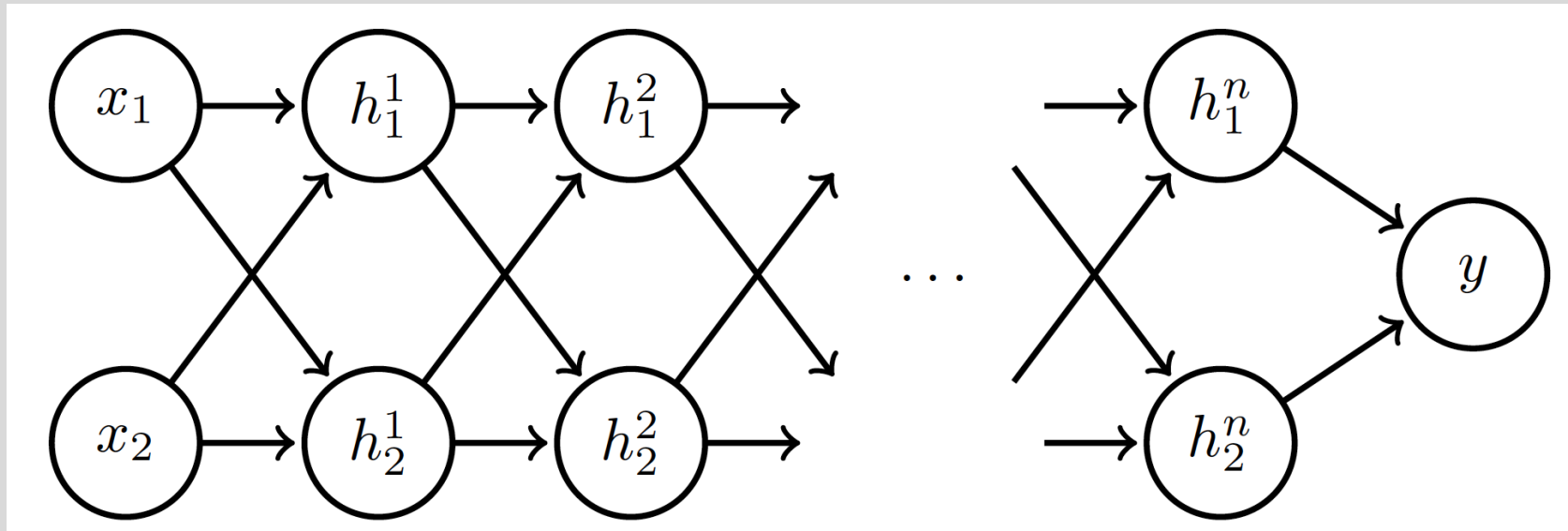


Image Source: deeplearning.ai

Linear activation $h_i = Wx$
 $h_i = Wh_{i-1}, \quad i = 2, \dots, n$



Exploding and Vanishing Gradients

Suppose $\mathbf{W} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$:

$$\begin{bmatrix} h_1^1 \\ h_2^1 \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \dots \quad \begin{bmatrix} h_1^n \\ h_2^n \end{bmatrix} = \begin{bmatrix} a^n & 0 \\ 0 & b^n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$



Exploding and Vanishing Gradients

Suppose $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Case 1: $a = 1, b = 2$:

$$y \rightarrow 1, \quad \nabla y \rightarrow \begin{bmatrix} n \\ n2^{n-1} \end{bmatrix} \quad \text{Explodes!}$$

Case 2: $a = 0.5, b = 0.9$:

$$y \rightarrow 0, \quad \nabla y \rightarrow \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{Vanishes!}$$



Exploding and Vanishing Gradients

Exploding gradients lead to cliffs

Can be mitigated using [gradient clipping](#)

if $\|g\| > u$

$$g \leftarrow \frac{gu}{\|g\|}$$

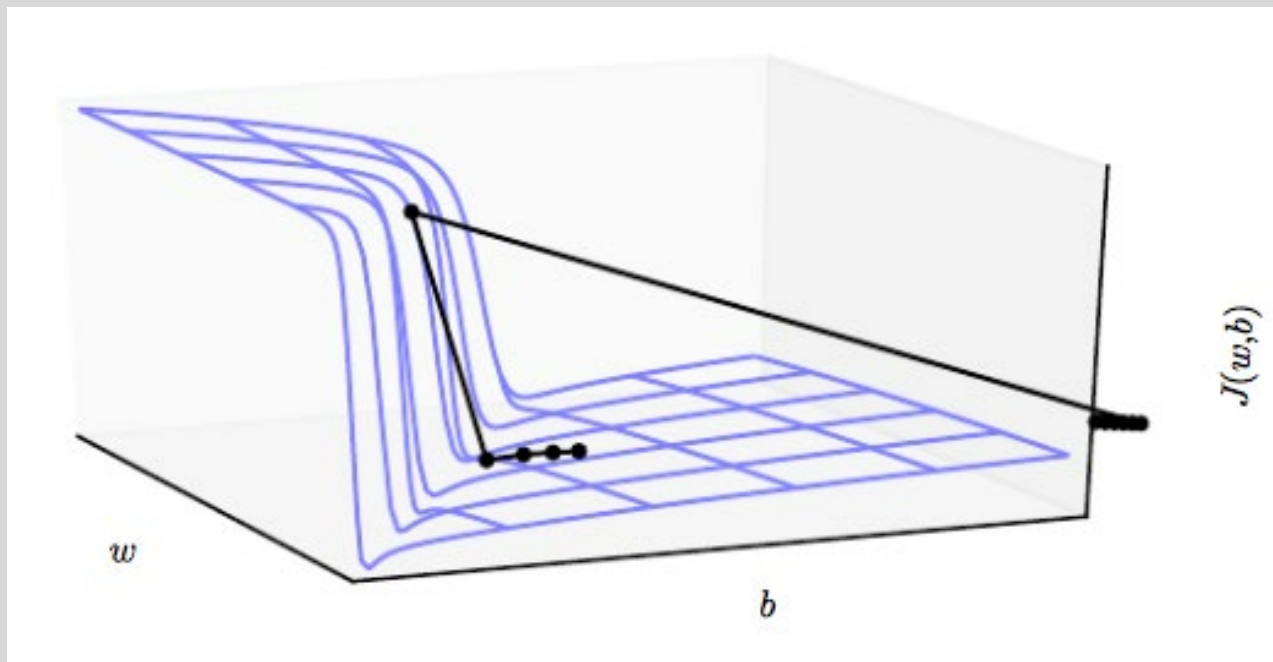


Image Source: Goodfellow et al. (2016)